YUTAO ZHANG 张聿韬

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Research & Technical Summary

PhD candidate in Computer Graphics at Université de Montréal, specializing in geometry processing, neural representations, and variational optimization. Experienced integrating geometry- and physics-inspired operators (curvature, Poisson/Laplacian constraints, FFT/spectral transforms) into neural network pipelines for 3D geometric tasks. Work accepted at SIGGRAPH Asia 2025.

Education

PhD in Computer Graphics 2023–Present
Université de Montréal GPA: **4.15/4.30**

BSc in Computer Science 2020–2022
Université de Montréal GPA: **4.24/4.30**
Excellence Scholarships: UdeM (2021, 2022), American Express (2021)

BA in French 2016–2020
Peking University GPA: **3.56/4.00**

Research Experience

Variational Neural Surfacing of 3D Sketches SIGGRAPH Asia 2025 (Accepted)

- Reformulated and reparameterized the flow-alignment energy for implicit surface representations.
- Developed a gradient-domain optimization method for flow-aligned surfacing from 3D sketches and clean curve networks.

Keywords: Neural Implicit Surfacing, Variational Optimization, 3D Sketch Processing.

Surfacing 3D Sketched Curve Networks with Minimal Currents 2022–2024

- Investigated minimal surface reconstruction for incomplete or noisy curve networks.
- Applied geodesic measure theory with spatial metric control.

Keywords: Minimal Surfaces, Curve Networks, Differential Geometry.

Expertise Breadth

Geometry Processing: Surface reconstruction, mesh analysis, subdivision, deformation.

Machine Learning: Representative learning, reinforcement learning, neural geometric processing, natural language processing, probabilistic graphical models.

Rendering & Image Synthesis: Ray tracing, shading, texture filtering, anti-aliasing.

Technical Skills

Programming: Python (proficient);
C++, Java, JavaScript, HTML (familiar); Haskell, Prolog (exposure).
Machine Learning: PyTorch (familiar); JAX (exposure).
Graphics & Geometry: Houdini (familiar); WebGL, Blender (exposure).
Mathematics: Differential geometry; optimization; probability theory.

Course Projects

Screened Poisson Surface Reconstruction (Python Implementation)

- Built a simplified screened Poisson reconstruction for uniform point samples.
- Added smoothing regularization for stable surface recovery.

Second-Order EWA Texture Filtering

- Improved elliptical weighted average texture mapping.
- Corrected sampling center after perspective projection using second-order derivatives.

Publications

Zhang, Y., Wang, S., Bessmeltsev, M. *Variational Neural Surfacing of 3D Sketches*. SIGGRAPH Asia 2025 (Accepted).

Industry & Internship Experience

Sinopec Engineering Group (SEG) — Information Center Intern 2019

- Learned enterprise data workflows using SAP BW and Oracle databases.
- Developed and maintained an internal corporate reporting website.

Foreign Language Teaching & Research Press — Editor/Proofreader 2018

- Edited and proofread French dictionary software content (text and audio).

Languages

Chinese (Native) | English (TOEFL 103) | French (B2) | Japanese (nearly N3)

Extracurricular

Peking University, School of Foreign Languages

- Basketball Team Member.
- Student Union Activities (Public Relations and Mini-program Development).

Interests

Video games: 20,000+ hours of gaming experience.
Music: Singing; violin.
Sports: Basketball; billiards; tennis; skiing.